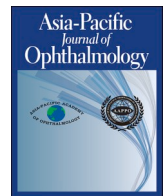




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Using artificial intelligence to detect keratoconus progression based on age and anterior segment optical coherence tomography images

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ABSTRACT

In this study, we aimed to establish and evaluate an index to define keratoconus progression based on map images obtained from anterior segment optical coherence tomography (AS-OCT) and age through deep learning. AS-OCT images and patient data were retrospectively retrieved from Hospital records. Classifiers for each of the six types of images were created using machine learning, and an index was established for judging keratoconus progression from the output adjusted or unadjusted for age. A total dataset of 2006 paired AS-OCT images from longitudinal examinations (positive: 883, negative: 1123), comprising a learning dataset of 1678 pairs (positive: 741, negative: 937) and a validation dataset of 328 pairs (positive: 142, negative: 186), was used. The final numerical value (index) was obtained from the output results and age. The index was subsequently evaluated using the validation dataset to determine its accuracy, sensitivity, specificity, goodness of fit, and area under the receiver operating characteristic curve (AUC). The average value of the keratoconus progression index significantly differed between the age-adjusted and unadjusted algorithms ($p < 0.001$). Adjusting for age improved the keratoconus progression index (accuracy, 0.655 vs. 0.909; sensitivity, 0.915 vs. 0.937; specificity, 0.457 vs. 0.887; precision, 0.563 vs. 0.864; F-value, 0.697 vs. 0.899; and AUC 0.727 vs. 0.935). Using machine learning on six types of AS-OCT images adjusted for age, we developed an index with a better predictive value of keratoconus progression than previously reported. This index accurately predicts keratoconus progression, which is useful for determining indications for corneal cross-linking using initial AS-OCT imaging.

1. Introduction

Keratoconus is an ocular disease characterized by progressive corneal ectasia leading to irregular astigmatism and requiring rigid gas-permeable lenses to correct vision. With progression, acute edema may develop at times, and contact lenses can no longer be worn, requiring corneal transplantation for vision improvement.^{17,23,26,34,8} The risk factors for keratoconus include eye rubbing, family history of keratoconus, and parental consanguinity.⁴ Globally, keratoconus ranks the second-most common indication for corneal transplantation after Fuchs' corneal endothelial dystrophy, accounting for approximately 30% of corneal transplantations.¹⁰ Recently, a corneal cross-linking (CXL) treatment using riboflavin eye drops and UV-A irradiation was

developed.³⁷ Other treatment options include surgical techniques, such as transepithelial CXL, accelerated CXL, and accelerated transepithelial CXL, which are increasingly being used.^{19,2,36} Notably, these CXL procedures reportedly reduce the requirement for corneal transplantation in these patients.²⁹ However, it is essential to detect disease progression when determining the indication for CXL surgery and evaluating its effectiveness in halting the progression. The progression of keratoconus is often determined by an increase in the maximum keratometry value (Kmax) obtained using the Scheimpflug camera or anterior segment optical coherence tomography (AS-OCT).^{7,9} Thus, repeated periodic examination is often required to detect keratoconus progression.

Deep learning has been used to develop various diagnostic indices^{18, 22,35} and applied to diagnosing keratoconus through discriminant

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analysis (^{1,16,20,21,27,28,31–33,38,5,6,12,14}). We have previously reported that the indices obtained from AS-OCT images could capture staging and progression using deep learning with an accuracy of 0.794.¹⁵ Accurate prediction of keratoconus progression has important implications not only for patients but also for clinicians and society. For patients, early identification of progression enables timely CXL and better visual outcomes. For clinicians, predictive models can support decision-making on CXL indication and follow-up strategies. From a societal perspective, preventing advanced disease helps reduce the need for corneal transplantation and alleviates associated healthcare and resource burdens. In the present study, we aimed to examine and evaluate a more appropriate index for determining the progression of keratoconus by incorporating patient age data into six types of maps obtained through AS-OCT to achieve greater accuracy.

2. Methods

2.1. Ethics consideration

This study was approved by the Research Ethics Committee of Miyata Eye Hospital (approval number: CS-315) and the Research Ethics Committee of the Faculty of Medicine of the University of Tokyo (2019354NI). The study protocol was registered in the University Hospital Medical Information Network Clinical Trial Registry (000040308). This study was conducted in accordance with the principles of the Declaration of Helsinki. Informed consent was obtained using an opt-out method, which allowed participants not to participate in this research, according to the permission of both research ethics committees.

2.2. Study population

This study included patients who were diagnosed with keratoconus by the corneal specialists at Miyata Eye Hospital and the University of Tokyo Hospital between October 2014 and November 2021 and followed up for more than 1 year using AS-OCT. To exclude the effect of

wearing contact lenses, patients who wore rigid gas-permeable and soft contact lenses were instructed to stop wearing them for 3 and 2 weeks, respectively, before this evaluation.^{24,3} To avoid data leakage, the dataset was split at the patient level rather than at the image level. Therefore, no patients were shared between the training and validation datasets. Multiple AS-OCT scans obtained at different time points from the same patient were included within each dataset to capture longitudinal changes. In the training dataset, patients contributed between 1 and 26 AS-OCT scans (median, 2; mean, 5.05; mode, 1). In the validation dataset, patients contributed between 1 and 28 AS-OCT scans (median, 1; mean, 3.90; mode, 1).

2.3. Definition of progression

Progression of keratoconus was defined as a ≥ 1.00 D increase in the maximum anterior curvature on corneal topography or worse corrected visual acuity accompanied by a ≥ 1.00 D increase in astigmatism, confirmed in two or more examinations over 12 months.

2.4. AS-OCT image acquisition

Imaging was performed using swept-source OCT SS-2000, CASIA 2 (Tomey Corporation, Nagoya, Japan). This model had an axial resolution of 10 μm , a transverse resolution of 30 μm , and a scan rate of 50000 A-scans/s. Six maps consisting of the anterior elevation (Elevation [Anterior]: EA), anterior curvature (axial power [Keratometry]: APK), posterior elevation (Elevation [Posterior]: EP), posterior curvature (axial power [Posterior]: APP), refractive power (Refractive Power [Keratometry]: RP), and pachymetry (Pachymetry: P) were used in this study (Fig. 1). An experienced examiner recorded AS-OCT images; only images with good-quality statements were used. Image quality was judged by the device's built-in quality index, exclusion of motion artifacts, and adequate visualization of the corneal apex.

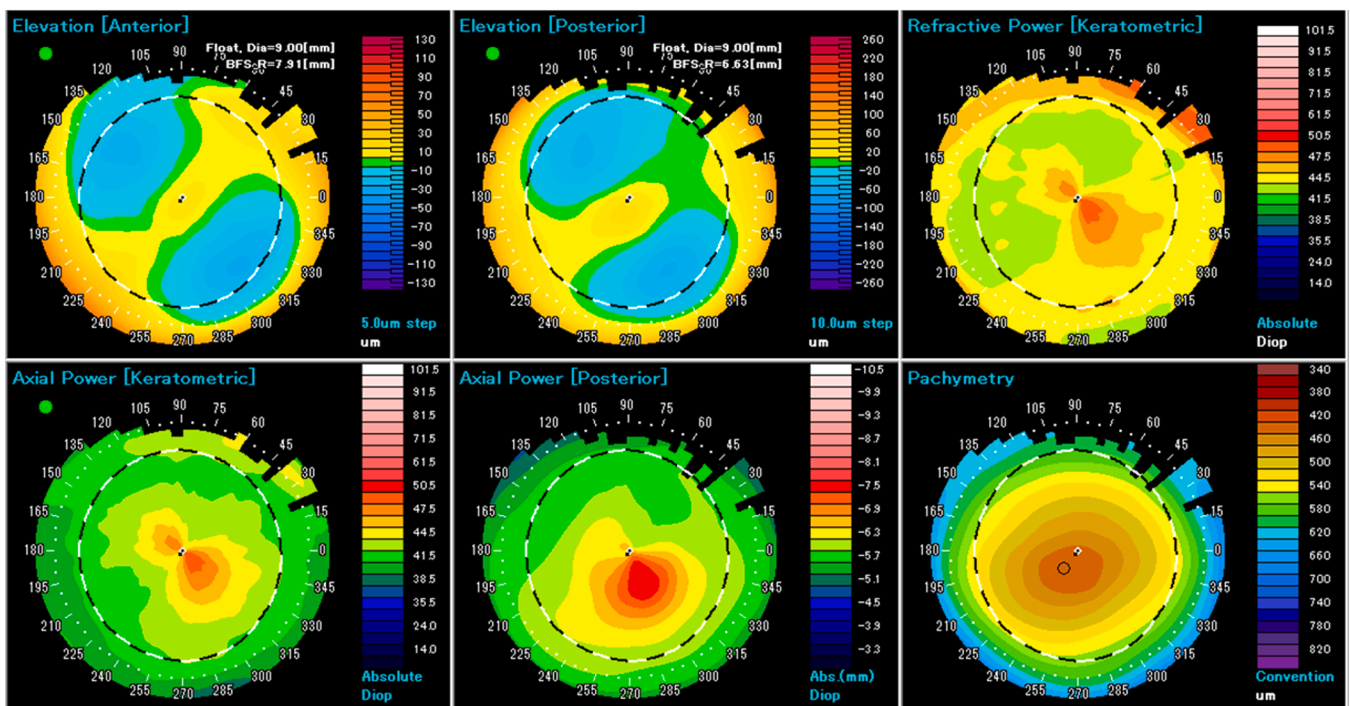


Fig. 1. A, representative image of six color-coded maps from anterior segment optical coherent tomography. The anterior elevation, posterior elevation, keratometry refractive power, keratometry axial power, posterior axial power, and pachymetry maps were selected.

2.5. Development of the keratoconus progression index using deep learning

The dataset consisted of 416 unique patients and was divided at the patient level into a training dataset ($n = 332$ patients) and a validation dataset ($n = 84$ patients), with no overlap between the two groups. Among the total dataset of 2006 pairs (positive: 883, negative: 1123), the learning dataset comprised 1678 pairs (positive: 741, negative: 937) and the validation dataset comprised 328 pairs (positive: 142, negative: 186). The number of scans per patient ranged from 1 to 26 (median: 2) in the training dataset and from 1 to 28 (median: 1) in the validation dataset. The data included observations of the same patient over different periods. Patient data were not shared between the training and validation datasets. The predicted rate of progression is converted to a number between 0.0 and 1.0, with 0.5 as the threshold. The network model was ResNet-50, chosen for its proven robustness and generalizability in medical imaging tasks including ophthalmology. Training parameters included a learning rate of 0.001, batch size of 32, 100 epochs, and the Adam optimizer. The EP classifier was pre-trained on EP+EA data to exploit the structural similarity between anterior and posterior corneal surfaces, thereby improving feature extraction. Five other classifiers were pre-trained on all data and fine-tuned on respective data.

Data augmentation involved the following techniques: a 90% crop factor, a maximum rotation angle of 10° , and a median filter. Data were normalized with a mean of 0.50 and a standard deviation of 0.25. Using machine learning, a classifier was created for each of the six image types, and the final numerical value (index) was obtained from the output results and age. Age was incorporated post hoc as a covariate by adjusting classifier outputs using a regression model, rather than as a direct image input.

2.6. Statistical analysis

The resulting index was evaluated using a validation dataset, and the accuracy, sensitivity, and specificity were calculated. A receiver operating characteristic curve was generated, and the area under the curve (AUC) was calculated. The statistical analysis was performed using the bell curve in Excel (Social Survey Research Information Co., Ltd., Tokyo, Japan). Statistical significance was defined as $p < 0.05$.

3. Results

The mean age of the patients was 32.92 ± 14.71 years, and the percentage of female participants was 33.4%. [Supplementary Table 1](#) summarizes the demographic characteristics of the study population, including age distribution, gender ratio, and regional distribution for both the training and validation datasets. In the training dataset, 741 scans (44.2%) were classified as progressive and 937 scans (55.8%) as stable. In the validation dataset, 142 scans (43.3%) were progressive and 186 scans (56.7%) were stable, indicating a comparable class distribution between datasets. The average value of the keratoconus progression index was 0.736 ± 0.230 (age-unadjusted) and 0.492 ± 0.299 (age-adjusted). The average value of the keratoconus progression index of data diagnosed as “progressive” was 0.840 ± 0.180 (age-unadjusted) and 0.737 ± 0.183 (age-adjusted) and that of data diagnosed as “non-progressive” was 0.657 ± 0.233 (age-unadjusted) and 0.233 ± 0.180 (age-adjusted). The average value of the keratoconus progression index significantly differed between the age-adjusted and -unadjusted algorithms ($p < 0.001$) ([Table 1](#)). The true positive rate was 93.7% and the true negative rate was 89.7% ([Table 2](#)). [Table 3](#) presents the sensitivity and specificity when changing the threshold of the keratoconus progression index. A threshold of 0.50 was selected as the optimal cutoff based on ROC analysis and Youden’s Index, which maximized the balance of sensitivity and specificity ([Table 4](#)). [Fig. 2](#) shows a histogram of positive scores for each classifier (all ages). [Supplemental Figs 1–5](#)

Table 1

Age-adjusted and -unadjusted keratoconus progression indices in the validation dataset.

	Without adjustment for age	With adjustment for age	P value
Progressive	0.840 ± 0.180	0.737 ± 0.183	< 0.001
Non-progressive	0.657 ± 0.233	0.233 ± 0.180	< 0.001
Total	0.736 ± 0.230	0.492 ± 0.299	< 0.001

Average $\pm$ SD, paired *t*-test

Table 2

Test results of the validation datasets.

	Estimation by index	
	Positive	Negative
Progressive	133 (True positive rate: 93.7%)	9 (False negative rate: 6.3%)
Non-progressive	21 (False positive rate: 11.3%)	165 (True negative rate: 89.7%)

illustrate the histograms of positive scores by age for each classifier.

The evaluation indices at the optimal threshold are listed in [Table 4](#). The results of the keratoconus progression index, without adjustment for age, had an accuracy of 0.655, a sensitivity of 0.915, a specificity of 0.457, a precision of 0.563, an F-measure of 0.697, and an AUC of 0.727. Adjusting for age improved all indices to an accuracy of 0.909, a sensitivity of 0.937, a specificity of 0.887, a precision of 0.864, an F-measure of 0.899, and an AUC of 0.935. [Fig. 3](#) depicts the AUC of each classifier. The detailed performance metrics of the individual classifiers, including accuracy, sensitivity, specificity, precision, and AUC, are provided in [Supplementary Table 2](#). [Fig. 4](#) presents the average AUC of all classifiers and the age-adjusted keratoconus index. To improve interpretability of the model, we performed explainability analysis using SHapley Additive exPlanations (SHAP). The SHAP summary plot demonstrated that patient age was most influential feature, followed by multiple AS-OCT-derived parameters, including pachymetry and posterior elevation, which contributed consistently to the keratoconus progression index. These findings support the complementary role of imaging features in predicting keratoconus progression ([Fig. 5](#)).

4. Discussion

Machine learning has been applied to images acquired from corneal topography, AS-OCT, or Scheimpflug imaging for diagnostic purposes in keratoconus.^{1,12,14,16,20,21,27,28,31–33,38,5,6} However, most studies on keratoconus have focused on diagnosis rather than progression prediction^{13,15,25,30}. Previous reports on AI-based approaches for detecting keratoconus progression are summarized in [Table 5](#). Importantly, in the present study, we demonstrated explainability of the keratoconus progression index model using SHAP ([Fig. 5](#)). The SHAP analysis quantified the relative contribution of each input, showing that pachymetry and posterior elevation were the strongest determinants of the prediction, followed by patient age. Age is a well-established risk factor for keratoconus progression and demonstrated a strong contribution in our model. Because the relationship between age and progression risk is largely monotonic, an age-only model would primarily reflect this known clinical trend and be strongly influenced by the age distribution of the dataset. In contrast, our analysis focused on evaluating the additional contribution of AS-OCT-derived features, which were shown to provide complementary predictive information. This suggests that AS-OCT-derived features provide incremental predictive value beyond age alone. These analyses enhance the clinical interpretability of the model and address the limitation of black-box AI approaches, thereby facilitating confidence in clinical decision-making.

CXL is increasingly being used to halt the progression of keratoconus.

Table 3

Sensitivity and specificity when changing the threshold of the keratoconus progression index.

Threshold	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45
TP/FN	142/0	142/0	142/0	142/0	142/0	142/0	142/0	142/0	133/9
Sensitivity	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.937
TN/FP	0/186	0/186	0/186	0/186	0/186	0/186	0/186	0/186	66/120
Specificity	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.355
0.50	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95
130/12	124/18	121/12	119/23	111/31	103/39	99/43	92/50	82/60	56/86
0.915	0.873	0.852	0.838	0.782	0.725	0.697	0.648	0.577	0.394
85/101	96/90	101/85	105/81	111/75	114/72	122/64	124/62	130/56	144/42
0.457	0.516	0.543	0.565	0.597	0.613	0.656	0.667	0.699	0.774

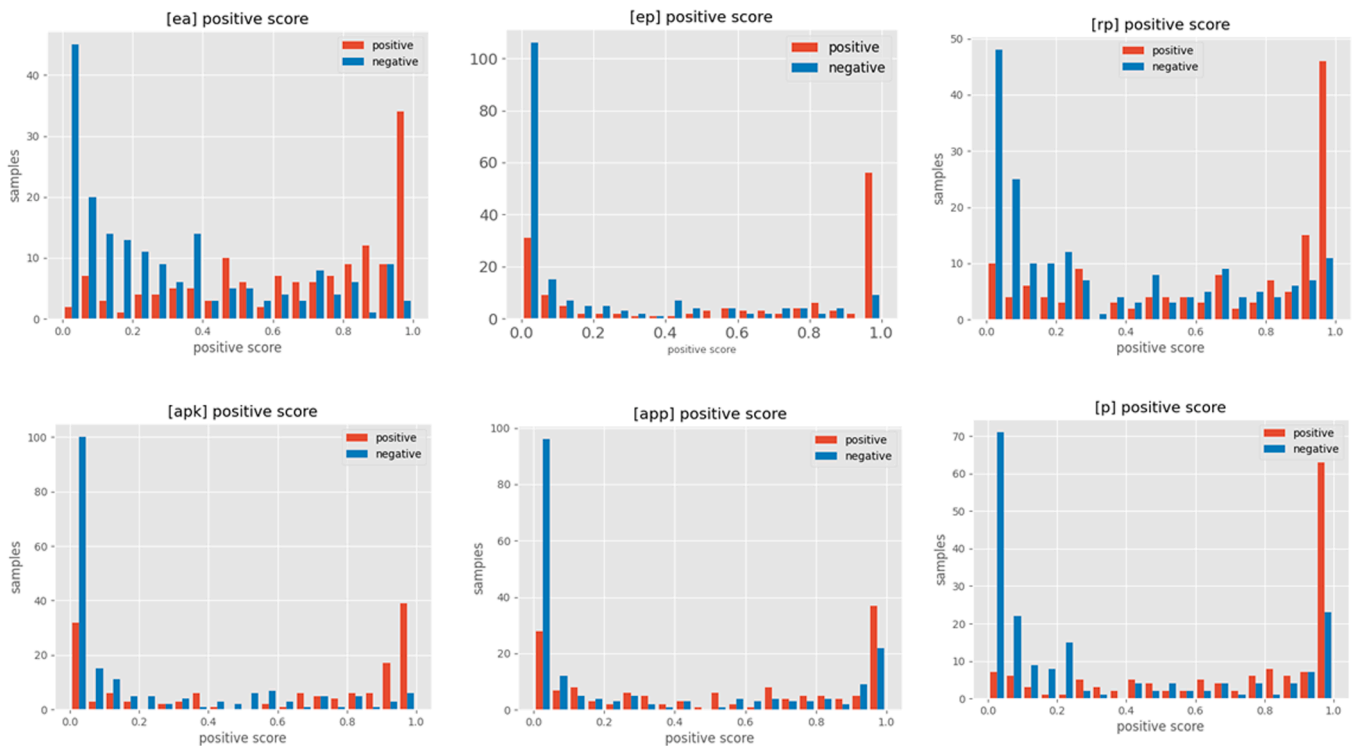
TP, True positive; FN, False negative; TN, True negative; FP, False positive.

Table 4

Evaluation index at the optimal threshold.

	Accuracy	Sensitivity	Specificity	Precision	F-measure	AUC
Without adjustment for age	0.655	0.915	0.457	0.563	0.697	0.727
With adjustment for age	0.909	0.937	0.887	0.864	0.899	0.935

Index > 0.5 was deemed as 'positive.' AUC: Area under the curve

**Fig. 2.** Histogram of positive scores for each classifier (all ages). ea: Elevation [Anterior], ep: Elevation [Posterior], rp: Refractive Power [Keratometry], apk: Axial Power [Keratometry], app: Axial Power [Posterior], p: Pachymetry.

Therefore, a more accurate method for predicting progression will enhance early decision-making regarding CXL surgical interventions, thereby facilitating the accurate selection of patients who will benefit from treatment. Keratoconus is a time-dependent disease, and longitudinal modeling approaches, such as time-to-event or survival analysis, may provide additional insights into predicting progression risk over time. However, in the present study, we focused on scan-level classification to reflect a clinically relevant scenario in which clinicians assess progression status at individual visits using available imaging data. The present study incorporated age and six types of AS-OCT images, resulting in improved accuracy, sensitivity, and specificity for predicting keratoconus progression compared with previous studies. As shown in [Supplementary Table 2](#), the individual classifiers contributed differently

to the final keratoconus progression index, highlighting the importance of integrating multiple corneal features for robust prediction.

CNN: Convolutional neural network models, Kmax: Maximum keratometry value, RsF: Front surface steepest radius, BSFS: Best fit sphere front, RmB: Back surface mean radius, LOGIK, Logistic index for keratoconus.

In this study, we established an index to capture the progress of keratoconus and evaluated it using deep learning on six types of AS-OCT images and patient age. In our previous study, we established a progression index, using only six types of images with adjustments but without considering age.¹⁵ This index exhibited an accuracy of 84.9%, a sensitivity of 58.1%, and a specificity of 84.9%. However, in the present study, the new index demonstrated superior accuracy, sensitivity, and

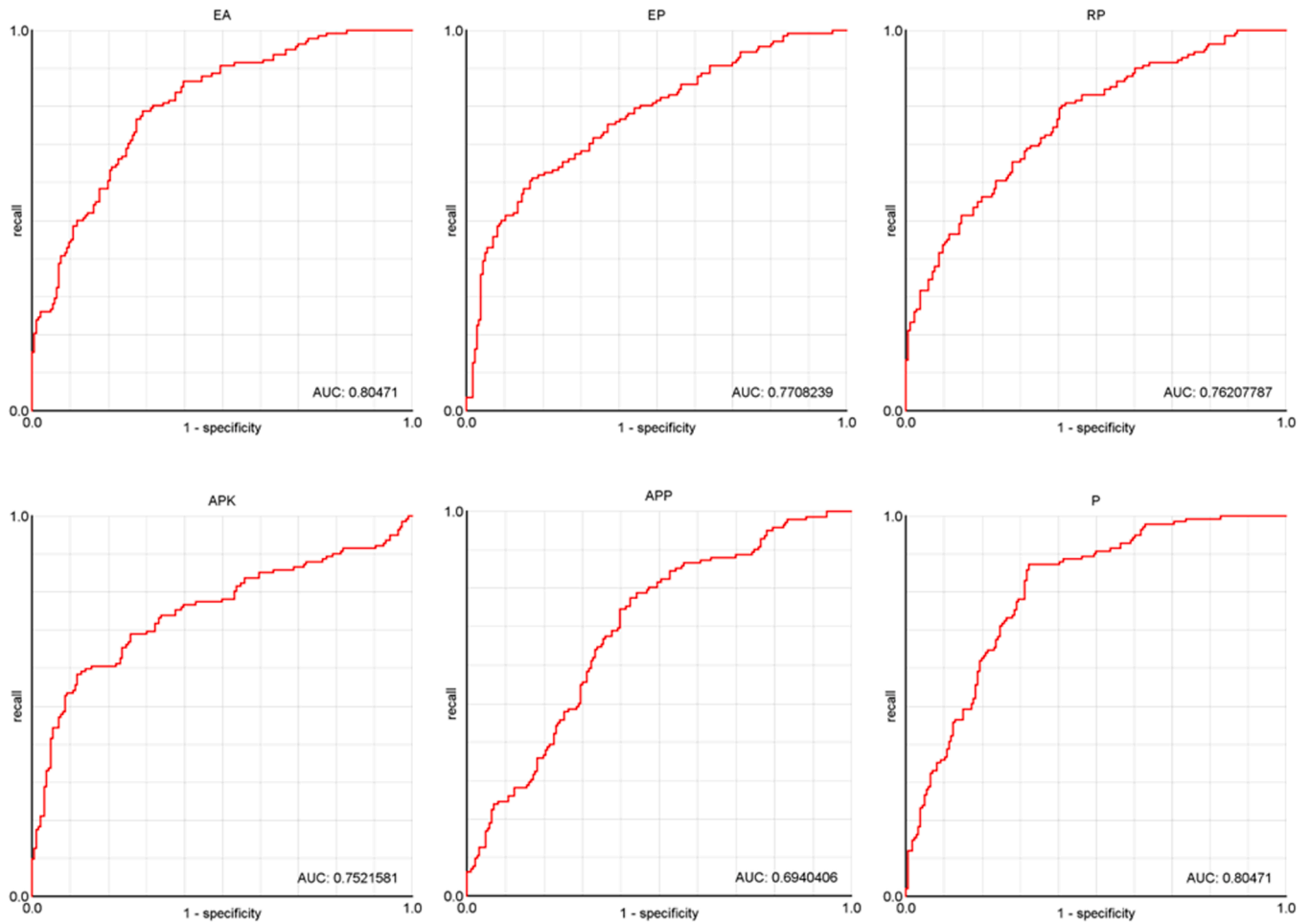


Fig. 3. Area under the receiver operating characteristic curves of the six classifiers. EA: Elevation [Anterior], EP: Elevation [Posterior], RP: Refractive Power [Keratometry], APK: Axial Power [Keratometry], APP: Axial Power [Posterior], P: Pachymetry, AUC: area under the curve.

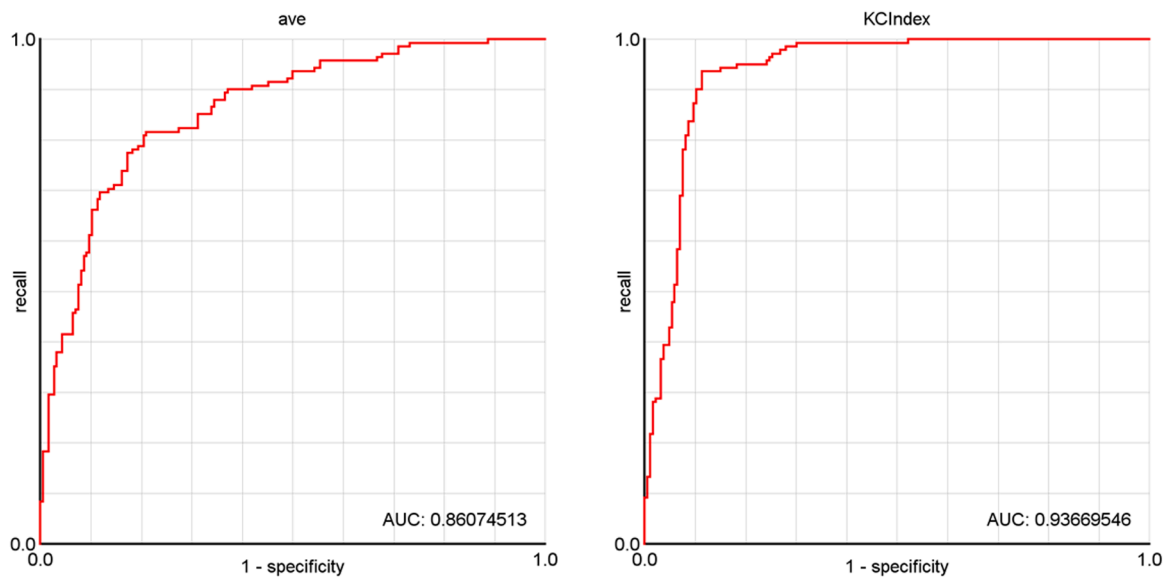


Fig. 4. Area under the receiver operating characteristic curve of the average of the six classifiers. Elevation [Anterior], Elevation [Posterior], Refractive Power [Keratometry], Axial Power [Keratometry], Axial Power [Posterior] and Pachymetry, and keratoconus progression index. ave: average of the six classifiers, KCIndex: keratoconus progression index, AUC: area under the curve.

specificity of 90.9%, 93.7%, and 88.7%, respectively, in the validation dataset.

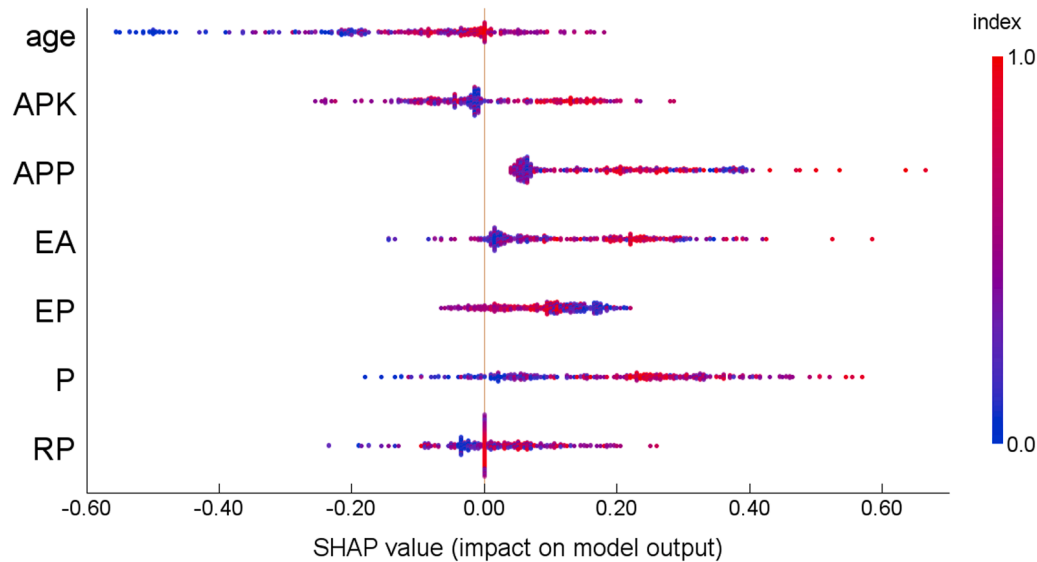


Fig. 5. SHAP summary (beeswarm) plot of feature importance in the prediction model. The SHAP summary plot illustrates the contribution of each feature to the model output across all samples. Each dot represents an individual scan, with the x-axis indicating the SHAP value (impact on model output). Positive SHAP values indicate increased likelihood of keratoconus progression, while negative values indicate decreased likelihood. Color represents the normalized feature value (blue: low, red: high). Patient age showed the largest contribution to model prediction, with younger age associated with a higher predicted probability of keratoconus progression. Anterior and posterior corneal parameters, including axial power and elevation metrics, also contributed to the model. These results demonstrate that, in addition to age, multiple AS-OCT-derived features contribute to model predictions. APK: Axial Power [Keratometry], APP: Axial Power [Posterior], EA: Elevation [Anterior], EP: Elevation [Posterior], P: Pachymetry, RP: Refractive Power [Keratometry].

Table 5

Previous reports on the diagnostic accuracy of artificial intelligence approaches for keratoconus progression.

Author	Year Journal	Sample size	Device	Input	Definition of progression	Sensitivity	Specificity	Accuracy	Precision
Reddy et al. (2022)	2022 EJO	385 eyes for model training and testing; 1331 eyes for prediction	Oculizer	4 maps (anterior elevation, posterior elevation, corneal thickness, and sagittal curvature)	Change in flat or steep K ≥ 1.0 D	-	-	Anterior elevation (97.46%), posterior elevation (84.81%), corneal thickness (83.54%), and sagittal curvature (71.15%)	-
Shetty et al. (2021)	2021 JRS	1518 follow-ups and 366 first visits	Pentacam	Total 31 parameters	Model A: an increase in Kmax by 0.75 D Model B: an increase in Kmax by 1.0 D Model C: an increase in Kmax by 1.25 D	Model A: 85% Model B: 86% Model C: 89%	Model A: 82% Model B: 82% Model C: 81%	Model A: 83% Model B: 88% Model C: 91%	Model A: 82.3% Model B: 87.8% Model C: 90.4%
Jimenez-Garcia et al. (2021)	2021 JCM	1155	Pentacam	6 features (age, KmaxZonalMean 3 mm, RsF, BFSF, RmB, and LOGIK)	significant changes in five parameters (KmaxZonalMean 3 mm, RsF, BFSF, RmB, or LOGIK) beyond a 95% confidence interval	70.8%	80.6%	-	71.4%
Kamiya et al. (2021)	2021 ATM	218	CASIA	6 maps and age-adjusted	Change in Kmax ≥ 1.0 D	95.5%	58.1%	84.9%	85.1%
This study		1778 (learning: 1460, validation: 328)	CASIA	6 maps and age-adjusted	Change in Kmax ≥ 1.0 D	94.2% 93.7%	41.9% 88.7%	79.4% 90.9%	80.3% 86.4%
				6 maps		91.5%	45.7%	65.5%	56.3%

In addition to the larger dataset, the incorporation of age yielded a clinically meaningful improvement in specificity. This enhances the clinical applicability of the index by enabling earlier and more reliable identification of patients at risk for progression, thereby supporting timely selection for cross-linking.

Despite our promising results, this study had several limitations. First, we used Kmax, a commonly used progression index for keratoconus. The global consensus on keratoconus defines progression as meeting two of the following three criteria: steepening of the anterior corneal surface, steepening of the posterior corneal surface, and/or

thinning or changes in the pachymetry rate of change.¹¹ Although the evaluation system uses a color-coded map to include these indicators, the results may vary depending on how the actual progression is defined. Second, wearing contact lenses may influence keratoconus, as rigid contact lenses are known to affect the corneal shape. The participants in this study were instructed to minimize the use of their contact lenses during the measurements; however, there is a possibility that full compliance was not achieved owing to lifestyle concerns. Third, although the dataset was split at the patient level to avoid data leakage, multiple scans from the same patient were included within each dataset. This may introduce intra-subject correlation, which could affect model generalizability. However, because the dataset was strictly split at the patient level, no data leakage occurred between the training and validation datasets. Moreover, in this study, using deep learning on six types of AS-OCT images and age data, we established an index that effectively captured the progression of keratoconus with high sensitivity and specificity in the validation dataset. Importantly, age was incorporated post hoc by adjusting classifier outputs with regression, rather than being directly input into the model, which may enhance generalizability. Furthermore, to enhance interpretability, we visualized the model's predictions using SHAP. Fig. 5 demonstrates that pachymetry and posterior elevation maps consistently contributed most to the keratoconus progression index, supporting the clinical relevance of these features in progression assessment. Although a more accurate prediction of the progression of keratoconus using a single image can facilitate the early selection of patients requiring CXL, further prospective studies at other centers are required to evaluate the performance of this index. This study was based on a dataset collected within a single country, and external validation using independent cohorts was not performed. Therefore, the generalizability of the model to other populations and clinical settings remains to be established.

In addition, the incorporation of SHAP analyses improved interpretability by demonstrating which image regions and features most strongly influenced predictions. This enhanced transparency may facilitate clinician acceptance and support the application of the keratoconus progression index in practice.

In conclusion, by using machine learning on six types of AS-OCT images and incorporating age adjustments, we developed a predictive index that significantly improves the accuracy of keratoconus progression assessment compared with previous methods. This age-adjusted index offers superior predictive value and enhances the ability to determine appropriate indications for CXL, thereby facilitating timely and effective intervention in clinical practice.

Author contributions

Conceived and designed the study: KK, YA, YK, and KM. Data collection: TM, HI, YM, and KM. Data analysis: KK, YA, YK, and NS. Drafting the manuscript: TM, KK, and YA. All authors reviewed the manuscript critically and approved the final version of the manuscript.

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Declaration of Competing Interest

Yuji Ayatsuka and Yudai Kato are employed by Cresco Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.apjo.2026.100323](https://doi.org/10.1016/j.apjo.2026.100323).

Data availability

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

References

- Accardo PA, Pensiero S. Neural network-based system for early keratoconus detection from corneal topography. *J Biomed Inf.* 2002;35(3):151–159. [https://doi.org/10.1016/s1532-0464\(02\)00513-0](https://doi.org/10.1016/s1532-0464(02)00513-0).
- Aixinjueluo W, Usui T, Miyai T, Toyono T, Sakisaka T, Yamagami S. Accelerated transepithelial corneal cross-linking for progressive keratoconus: a prospective study of 12 months. *Br J Ophthalmol.* 2017;101(9):1244–1249. <https://doi.org/10.1136/bjophthalmol-2016-309775>.
- Akiyama K, Ono T, Ishii H, et al. Impact of rigid gas-permeable contact lens on keratometric indices and corneal thickness of keratoconus eyes examined with anterior segment optical coherence tomography. *PLOS One.* 2022;17(7), e0270519. <https://doi.org/10.1371/journal.pone.0270519>.
- Almusawi LA, Hamied FM. Risk factors for development of keratoconus: a matched pair case-control study. *Clin Ophthalmol.* 2021;15:3473–3479. <https://doi.org/10.2147/OPTH.S248724>.
- Arbelaez MC, Versaci F, Vestri G, Barboni P, Savini G. Use of a support vector machine for keratoconus and subclinical keratoconus detection by topographic and tomographic data. *Ophthalmology.* 2012;119(11):2231–2238. <https://doi.org/10.1016/j.ophtha.2012.06.005>.
- Dos Santos VA, Schmetterer L, Stegmann H, et al. CorneaNet: fast segmentation of cornea OCT scans of healthy and keratoconic eyes using deep learning. *Biomed Opt Express.* 2019;10(2):622–641. <https://doi.org/10.1364/BOE.10.000622>.
- Epstein RL, Chiu YL, Epstein GL. Pentacam HR criteria for curvature change in keratoconus and postoperative LASIK ectasia. *J Refract Surg.* 2012;28(12):890–894. <https://doi.org/10.3928/1081597X-20121115-04>.
- Fan Gaskin JC, Patel DV, McGhee CN. Acute corneal hydrops in keratoconus - new perspectives. *Am J Ophthalmol.* 2014;157(5):921–928. <https://doi.org/10.1016/j.ajo.2014.01.017>.
- Fujimoto H, Maeda N, Shintani A, et al. Quantitative evaluation of the natural progression of keratoconus using three-dimensional optical coherence tomography. *OCT169-75 Invest Ophthalmol Vis Sci.* 2016;57(9). <https://doi.org/10.1167/iovs.15-18833>.
- Gain P, Jullienne R, He Z, et al. Global survey of corneal transplantation and eye banking. *JAMA Ophthalmol.* 2016;134(2):167–173. <https://doi.org/10.1001/jamaophthalmol.2015.4776>.
- Gomes JA, Tan D, Rapuano CJ, et al. Global consensus on keratoconus and ectatic diseases. *Cornea.* 2015;34(4):359–369. <https://doi.org/10.1097/ICO.0000000000000408>.
- Issarti I, Consejo A, Jiménez-García M, Hershko S, Koppen C, Rozema JJ. Computer aided diagnosis for suspect keratoconus detection. *Comput Biol Med.* 2019;109:33–42. <https://doi.org/10.1016/j.combiomed.2019.04.024>.
- Jiménez-García M, Issarti I, Kreps EO, et al. Forecasting progressive trends in keratoconus by means of a time delay neural network. *J Clin Med.* 2021;10(15):3238. <https://doi.org/10.3390/jcm10153238>.
- Kamiya K, Ayatsuka Y, Kato Y, et al. Keratoconus detection using deep learning of colour-coded maps with anterior segment optical coherence tomography: a diagnostic accuracy study. *BMJ Open.* 2019;9(9), e031313. <https://doi.org/10.1136/bmjopen-2019-031313>.
- Kamiya K, Ayatsuka Y, Kato Y, et al. Prediction of keratoconus progression using deep learning of anterior segment optical coherence tomography maps. *Ann Transl Med.* 2021;9(16):1287. <https://doi.org/10.21037/atm-21-1772>.
- Kovács I, Miháľt K, Kránitz K, et al. Accuracy of machine learning classifiers using bilateral data from a Scheimpflug camera for identifying eyes with preclinical signs of keratoconus. *J Cataract Refract Surg.* 2016;42(2):275–283. <https://doi.org/10.1016/j.jcrs.2015.09.020>.
- Krachmer JH, Feder RS, Belin MW. Keratoconus and related noninflammatory corneal thinning disorders. *Surv Ophthalmol.* 1984;28(4):293–322. [https://doi.org/10.1016/0039-6257\(84\)90094-8](https://doi.org/10.1016/0039-6257(84)90094-8).
- Le Callet P, Viard-Gaudin C, Barba D. A convolutional neural network approach for objective video quality assessment. *IEEE Trans Neural Netw.* 2006;17(5):1316–1327. <https://doi.org/10.1109/TNN.2006.879766>.
- Leccisotti A, Islam T. Transepithelial corneal collagen cross-linking in keratoconus. *J Refract Surg.* 2010;26(12):942–948. <https://doi.org/10.3928/1081597X-20100212-09>.
- Maeda N, Klyce SD, Smolek MK. Neural network classification of corneal topography. Preliminary demonstration. *Investig Ophthalmol Vis Sci.* 1995;36(6):1327–1335.
- Maeda N, Klyce SD, Smolek MK, Thompson HW. Automated keratoconus screening with corneal topography analysis. *Invest Ophthalmol Vis Sci.* 1994;35(6):2749–2757.

22. Matsugu M, Mori K, Mitari Y, Kaneda Y. Subject independent facial expression recognition with robust face detection using a convolutional neural network. *Neural Netw.* 2003;16(5-6):555–559. [https://doi.org/10.1016/S0893-6080\(03\)00115-1](https://doi.org/10.1016/S0893-6080(03)00115-1).
23. Rabinowitz YS. Keratoconus. *Surv Ophthalmol.* 1998;42(4):297–319. [https://doi.org/10.1016/S0039-6257\(97\)00119-7](https://doi.org/10.1016/S0039-6257(97)00119-7).
24. Rao SK, Fogla R, Padmanabhan P, Sitalakshmi G, Rajagopal R. Corneal warpage after discontinuation of contact lens wear. *Cornea.* 2000;19(2):207–210. <https://doi.org/10.1097/00003226-200003000-00015>.
25. Reddy JC, Bhamidipati P, Dwivedi S, et al. KEDOP: keratoconus early detection of progression using tomography images. *Eur J Ophthalmol.* 2022;32(5):2554–2564. <https://doi.org/10.1177/11206721221087566>.
26. Romero-Jiménez M, Santodomingo-Rubido J, Wolffsohn JS. Keratoconus: a review. *Cont Lens Anterior Eye.* 2010;33(4):157–166. <https://doi.org/10.1016/j.clae.2010.04.006>.
27. Ruiz Hidalgo I, Rodriguez P, Rozema JJ, et al. Evaluation of a machine-learning classifier for keratoconus detection based on scheimpflug tomography. *Cornea.* 2016;35(6):827–832. <https://doi.org/10.1097/ICO.0000000000000834>.
28. Ruiz Hidalgo I, Rozema JJ, Saad A, et al. Validation of an objective keratoconus detection system implemented in a scheimpflug tomographer and comparison with other methods. *Cornea.* 2017;36(6):689–695. <https://doi.org/10.1097/ICO.0000000000001194>.
29. Sandvik GF, Thorsrud A, Råen M, Østern AE, Sæthre M, Drolsum L. Does corneal collagen cross-linking reduce the need for keratoplasties in patients with keratoconus. *Cornea.* 2015;34(9):991–995. <https://doi.org/10.1097/ICO.0000000000000460>.
30. Shetty R, Kundu G, Narasimhan R, et al. Artificial intelligence efficiently identifies regional differences in the progression of tomographic parameters of keratoconic corneas. *J Refract Surg.* 2021;37(4):240–248. <https://doi.org/10.3928/1081597X-20210120-01>.
31. Smadja D, Touboul D, Cohen A, et al. Detection of subclinical keratoconus using an automated decision tree classification. *Am J Ophthalmol.* 2013;156(2):237–246.e1. <https://doi.org/10.1016/j.ajo.2013.03.034>.
32. Smolek MK, Klyce SD. Current keratoconus detection methods compared with a neural network approach. *Investig Ophthalmol Vis Sci.* 1997;38(11):2290–2299.
33. Souza MB, Medeiros FW, Souza DB, Garcia R, Alves MR. Evaluation of machine learning classifiers in keratoconus detection from orbscan II examinations. *Clinics.* 2010;65(12):1223–1228. <https://doi.org/10.1590/S1807-59322010001200002>.
34. Suzuki M, Amano S, Honda N, Usui T, Yamagami S, Oshika T. Longitudinal changes in corneal irregular astigmatism and visual acuity in eyes with keratoconus. *Jpn J Ophthalmol.* 2007;51(4):265–269. <https://doi.org/10.1007/s10384-007-0453-2>.
35. Tivive FH, Bouzerdoum A. Efficient training algorithms for a class of shunting inhibitory convolutional neural networks. *IEEE Trans Neural Netw.* 2005;16(3):541–556. <https://doi.org/10.1109/TNN.2005.845144>.
36. Touboul D, Efron N, Smadja D, Praud D, Malet F, Colin J. Corneal confocal microscopy following conventional, transepithelial, and accelerated corneal collagen cross-linking procedures for keratoconus. *J Refract Surg.* 2012;28(11):769–776. <https://doi.org/10.3928/1081597X-20121016-01>.
37. Wollensak G, Spoerl E, Seiler T. Riboflavin/ultraviolet-a-induced collagen crosslinking for the treatment of keratoconus. *Am J Ophthalmol.* 2003;135(5):620–627. [https://doi.org/10.1016/S0002-9394\(02\)02220-1](https://doi.org/10.1016/S0002-9394(02)02220-1).
38. Yousefi S, Yousefi E, Takahashi H, et al. Keratoconus severity identification using unsupervised machine learning. *PLOS One.* 2018;13(10), e0205998. <https://doi.org/10.1371/journal.pone.0205998>.

Glossary

AS-OCT: Anterior segment optical coherence tomography
 AUC: Area under the receiver operating characteristic curve
 CXL: Corneal cross-linking
 UV-A: Ultraviolet A
 EA: Elevation [Anterior]
 APK: Axial power [Keratometry]
 EP: Elevation [Posterior]
 APP: Axial power [Posterior]
 RP: Refractive power [Keratometry]
 P: Pachymetry